

Tushar Agrawal

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EDUCATION

- The LNM Institute of Information Technology** Jaipur, India
Bachelor of Technology - **Computer Science**; CGPA: **7.63** July 2023 - June 2027
Courses: Object-Oriented Programming, Data Structures and Algorithms, Design and Analysis of Algorithms, Python Programming, Database Management Systems, Computer Organization and Architecture, Operating Systems, Computer Networks
- Rani Laxmibai Public School(ISC Board)** Jhansi, India
Class XII (Science Stream) – Percentage: **96.2%** 2023

EXPERIENCE

- Voice Games** Remote
Software Engineer (AI) Intern May 2026 - Present
 - Engineered Tracelens**, an AI-assisted frontend performance intelligence platform that automates browser audits, trace analysis, bottleneck detection, and optimization workflows for the core engineering team.
 - Optimized live gameplay latency** on the **Jest.com** platform using Tracelens to monitor, benchmark, and scale runtime metrics, directly accelerating asset-loading speeds and mitigating client-side rendering lag.
 - Built an automated analysis pipeline** integrating Playwright, Lighthouse, hydration analysis, and bundle intelligence into Tracelens for automated, real-time root-cause detection.
 - Developed a CLI-driven workflow** for multi-run audits, regression comparison, and consolidated performance reporting, standardizing the evaluation harness used by the team prior to production releases.

PROJECTS

- BlogSmith** | *React.js, Node.js, Express.js, MongoDB, JWT, Cloudinary* GitHub 
 - Built a **full-stack blogging platform** with end-to-end **CRUD operations**, integrating Cloudinary for efficient image storage and management.
 - Implemented secure authentication and session handling using **JWT** (access refresh tokens) and **bcrypt**, with protected routes and dynamic UI rendering.
 - Designed and integrated **RESTful APIs** with **MongoDB**, enabling seamless data flow across frontend, backend, and database layers.
- Codonova – Autonomous AI Development Framework** | *FastApi, React.js, Neo4j, ChromaDB, Docker* GitHub 
 - Built a **multi-agent AI framework** capable of autonomously planning, generating, testing, debugging, and self-correcting production-ready software codebases.
 - Designed a **graph-driven orchestration pipeline** using **Neo4j** and **ChromaDB** for dependency-aware task execution and persistent contextual memory.
 - Implemented an automated **TDD-based self-correction engine** with **Pytest/Jest**, improving overall code generation success rate from 76% to 96% through iterative debugging and retries.
 - Engineered fault-tolerant LLM infrastructure with API key rotation, exponential backoff, and provider fallback mechanisms for reliable uninterrupted execution.

SKILLS SUMMARY

- Languages:** C, C++, Javascript, Python
- Frameworks & Libraries:** React.js, Node.js, Next.js, Express.js, FastAPI
- Databases & Tools:** ChromaDB, Neo4j, MongoDB, Docker, Git/Github
- Concepts & Methodologies:** Agentic Workflows, Context Engineering, RAG, REST APIs, JWT, Structured Outputs
- Soft Skills:** Leadership, Event Management, Public Speaking

EXTRACURRICULAR

- General Secretary – Technology Council, LNMIIT** | *Event Management, Leadership* Aug 2025 – Present
 - Lead **7+ Science & technology clubs**, manage budgets and resources, and coordinate initiatives across diverse domains and campus-wide activities.
 - Oversee **PLINTH**, LNMIIT's 3-day national **technical fest**, heading a team of **100+ members** and driving strategic planning, execution, and cross-team collaboration for **40+ events** and **1,000+ participants**.